

Differentiation Rules

The Constant Rule:

$$\frac{d}{dx}(c) = 0$$

The Constant Multiple Rule:

$$\frac{d}{dx}[c \cdot f(x)] = c \cdot f'(x)$$

The Power Rule:

$$\frac{d}{dx}(x^n) = n \cdot x^{n-1}$$

The Sum Rule:

$$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$$

The Difference Rule:

$$\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$$

The Product Rule:

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

The Quotient Rule:

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - g'(x)f(x)}{[g(x)]^2}$$

Chain Rule:

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

Derivatives of Exponential Functions:

$$\frac{d}{dx}(b^x) = b^x \ln(b) \quad \text{which implies}$$

$$\frac{d}{dx}(e^x) = e^x$$

Derivatives of Logarithmic Functions:

$$\frac{d}{dx}[\log_b(x)] = \frac{1}{x \ln(b)} \quad \text{which implies}$$

$$\frac{d}{dx}[\ln(x)] = \frac{1}{x}$$

Derivatives of Trigonometric Functions:

$$\frac{d}{dx}[\sin(x)] = \cos(x)$$

$$\frac{d}{dx}[\cos(x)] = -\sin(x)$$

$$\frac{d}{dx}[\tan(x)] = \sec^2(x)$$

$$\frac{d}{dx}[\cot(x)] = -\csc^2(x)$$

$$\frac{d}{dx}[\sec(x)] = \sec(x)\tan(x)$$

$$\frac{d}{dx}[\csc(x)] = -\csc(x)\cot(x)$$

Derivatives of Inverse Trigonometric Functions:

$$\frac{d}{dx}[\sin^{-1}(x)] = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}[\cos^{-1}(x)] = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}[\tan^{-1}(x)] = \frac{1}{1+x^2}$$

$$\frac{d}{dx}[\cot^{-1}(x)] = \frac{-1}{1+x^2}$$

$$\frac{d}{dx}[\sec^{-1}(x)] = \frac{1}{|x|\sqrt{x^2-1}}$$

$$\frac{d}{dx}[\csc^{-1}(x)] = \frac{-1}{|x|\sqrt{x^2-1}}$$